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Paleomagnetic age of ferruginous weathering beneath the Hamersley Surface, Pilbara, Western Australia, and the Cenozoic apparent polar wander path

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ABSTRACT

During the Mesozoic and Paleogene, the Precambrian rocks in the Pilbara, Western Australia, underwent erosion and deep weathering that produced an undulating landform now represented by the duricrusted and partly eroded Hamersley Surface. A reddened, ferruginous weathering zone occurs immediately beneath this duricrusted surface. Oriented block samples of ferruginised strata of the Neoarchean–early Paleoproterozoic Hamersley Group exposed within approximately 15 m below the duricrust were collected at 20 sites in roadcuts along the Great Northern Highway between Munjina and Newman and exposures along the adjoining Karijini Drive. Stepwise thermal demagnetisation of cored specimens revealed a stable, high-temperature (680°C) component carried by hematite, with a mean direction ($n = 55$ specimens) of declination $D = 182.0°$, inclination $I = 52.9°$ ($\alpha_{95} = 3.6°$), indicating a pole position at latitude $\lambda_p = 77.6°$S, longitude $\varphi_p = 113.2°$E ($A_{95} = 4.3°$) and a paleolatitude $\lambda = 33.5 +3.6/–3.3°$S. Both normal and reversed polarities are present, indicating that the remanent magnetism was acquired over an interval of at least two polarity chrons (say 10^5–10^6 years). Chi-square tests on the determined pole position and three different sets of Cenozoic poles, namely those for dated volcanic rocks in eastern Australia supplemented by poles for Australian Cenozoic weathering horizons, and inferred poles from Pacific Ocean and Indian Ocean hotspot analyses and North American Cenozoic poles rotated to Australian coordinates, yielded a mean age of *ca* 24 ± 3 Ma, i.e. late Oligocene to early Miocene, interpreted as the time of formation of hematite in the sampled ferruginous zone. The ferruginous weathering occurred under globally warm conditions and was followed during the early to middle Miocene climatic optimum by the deposition of channel iron deposits, which incorporated detrital hematitic material derived from erosion of the ferruginous weathering zone beneath the Hamersley Surface.

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Pilbara Craton; Hamersley Province; Hamersley Surface; lateritic weathering; paleomagnetism; apparent polar wander path; Oligocene; Miocene; channel iron deposits

Introduction

Remnants of ancient landforms, ranging in age from Precambrian to Cenozoic, are recognised in Australia (Anand, 2005; Ollier, 1991, 2014; Twidale, 1994, 1997). These paleo-landforms commonly are mantled by deeply weathered rocks and regoliths, some of which are of major economic importance. In Western Australia, for example, economic deposits associated with weathering on paleo-landforms include bauxite in the Darling Range, Ni laterite at Bulong near Kalgoorlie, and Au at Boddington in the Darling Range and at Mount Gibson (Anand, 2005; Anand & Paine, 2002).

Methods of absolute dating of ancient landforms, their weathering zones and regoliths include radiometric techniques, organic and isotopic geochemistry, paleontology and paleomagnetism (Pillans, 2005, 2008; Watchman & Twidale, 2002). Following paleomagnetic dating of late Cenozoic lateritisation in the Perth Basin (Schmidt & Embleton, 1976), paleomagnetic studies in several continents have dated weathering zones and regoliths ranging in age from Cenozoic (Idnurm & Senior, 1978; Schmidt & Ollier, 1988; Schmidt, Prasad, & Raman, 1983; Théveniaut & Freyssinet, 1999) and Mesozoic (Ricordel, Parcerisa, Thiry, Moreau, & Gómez-Gras, 2007; Théveniaut, Quesnel, Wyns, & Hugues, 2007) to Proterozoic (Williams & Schmidt, 1997).

Here we apply paleomagnetism to determine the age of the ferruginous weathering zone immediately beneath the duricrusted Paleogene Hamersley Surface developed on Precambrian rocks in the Pilbara, Western Australia, and how this weathering zone relates to channel iron deposits (CID) along the paleo-Robe River and tributaries (Morris & Ramanaidou, 2007).

Geological setting

Precambrian

The 3.6–2.8 Ga (Paleo- to Mesoarchean) granite–greenstone terrane of the north Pilbara Craton (Figure 1) is unconformably overlain by the Mount Bruce Supergroup comprising the 2780–2629 Ma (Neoarchean) Fortescue Group of volcanic, intrusive igneous and sedimentary rocks, and the 2629–2449

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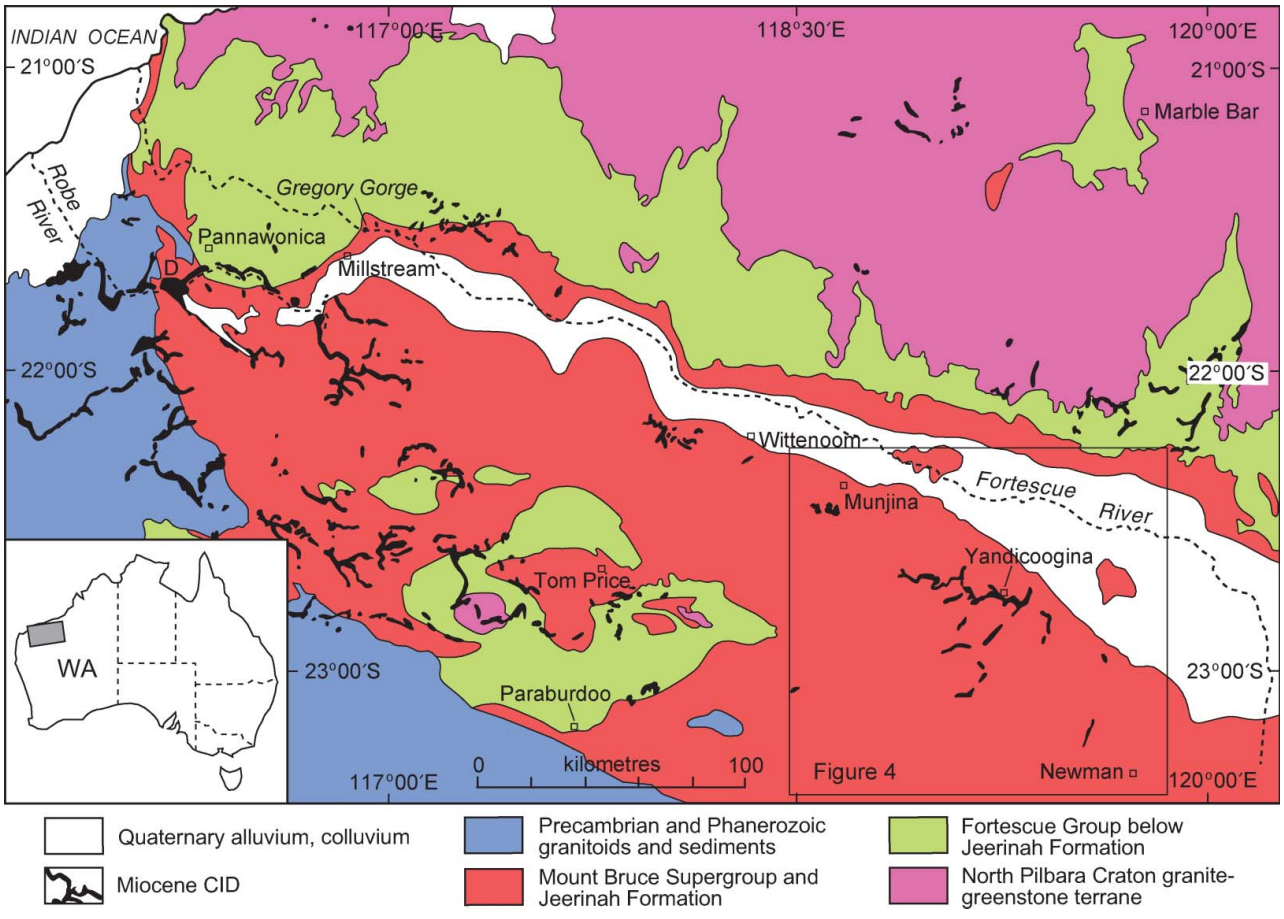


Figure 1. Simplified geological map of the north and west Pilbara Craton, showing the distribution of Precambrian rocks, exposed Miocene CID, and the study area. D, Deepdale. Geology adapted from Ramanaidou et al. (2003).

Ma (Neoarchean–early Paleoproterozoic) Hamersley Group of intrusive igneous and sedimentary rocks, including banded iron-formation (Griffin, 1990; Trendall, Compston, Nelson, De Laeter, & Bennett, 2004). The Fortescue and Hamersley groups in the southern Hamersley Basin underwent E–W to NW–SE folding during the Paleoproterozoic Ophthalmian deformation (Tyler & Thorne, 1990) prior to deposition of the overlying Paleoproterozoic Wyloo and Turee Creek groups. Iron ore is associated with the Neoarchean Marra Mamba Iron Formation and lower Paleoproterozoic Brockman Iron Formation (Blockley, 1990; Trendall, 1990; Trendall et al., 2004).

Mesozoic–Cenozoic

The Pilbara region underwent a protracted interval of erosion and deep weathering during the Mesozoic–Paleogene (Twidale, 1997), which produced an undulating landform that now has a partly eroded, ferruginous and siliceous duricrust (Figure 2). Remnants of a lateritic regolith with pedogenic hematitic pisoliths are preserved locally. Campana, Hughes, Burns, Whitcher, and Muceniekas (1964, p. 20) named this ancient landform the 'Hamersley Surface,' noting its relief of 450 m and more 'with smooth (though high) ridges and spurs, separated by mature valleys and broad plains…. This surface is now attacked and sharply carved by the modern cycle of erosion.'

Episodes of weathering and ferruginisation tens of metres deep recognised beneath the Hamersley Surface are estimated to have occurred in the early Eocene (HS1), middle Eocene (HS2), middle Miocene (HS3) and Pliocene (HS4) (Morris & Ramanaidou, 2007). Cenozoic iron deposits include Eocene and Pliocene hematite–goethite detrital iron deposits and Miocene hematite–goethite CID (Morris & Ramanaidou, 2007; Ramanaidou & Morris, 2010; Ramanaidou, Morris, & Horwitz, 2003). The HS3 weathering episode was intense, and the stripping of the then relatively easily eroded lateritic regolith beneath the Hamersley Surface was followed by deposition of extensive riverine CID, which attains a thickness of approximately 100 m and forms a series of mesas that rise above the level of the adjacent plain by inversion of relief over a distance of 150 km along the meandering course of the paleo-Robe River and other paleochannels (Figure 1) (Morris & Ramanaidou, 2007; Ramanaidou & Morris, 2010). The bulk of the CID is granular, comprising iron-rich granules, ooids, pisoids, peloids and ferruginised fossil wood, with goethite matrix and cement (Morris & Ramanaidou, 2007). CID are a major source of iron ore in the Hamersley Province, with typical grades of 57.4–59 wt% Fe (Ramanaidou et al., 2003).

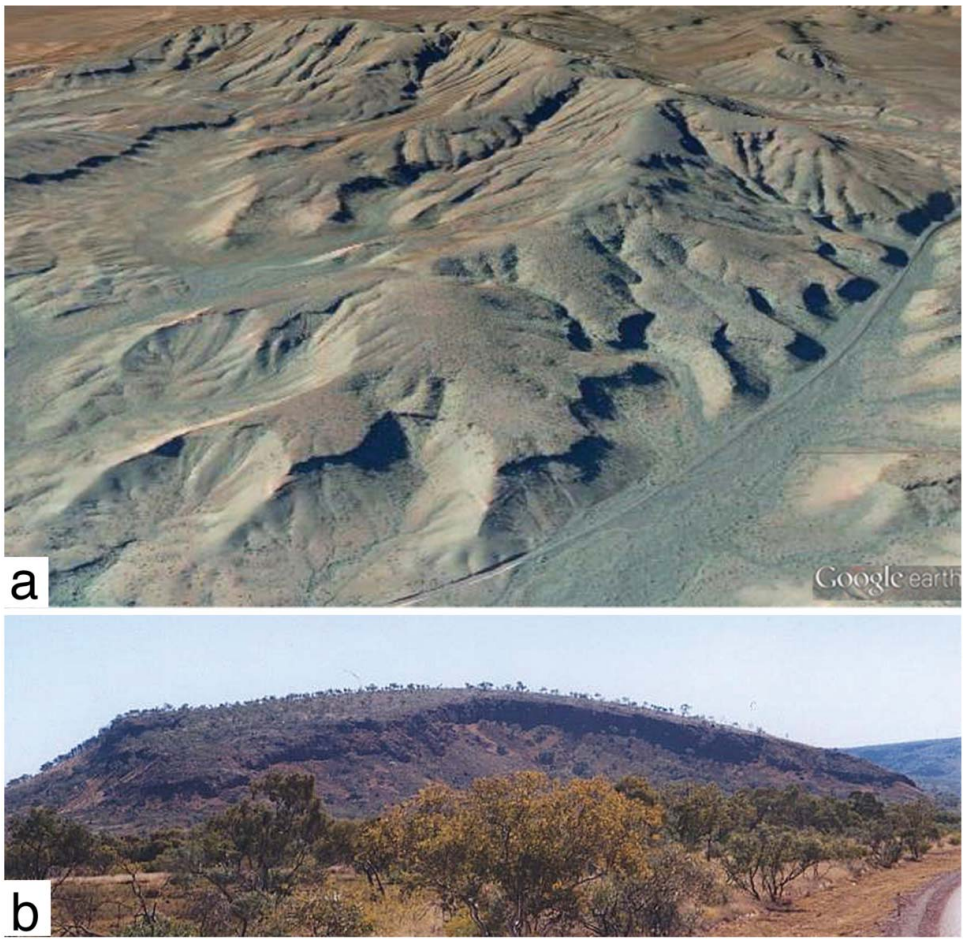


Figure 2. (a) Google Earth view looking ESE from 1300 m altitude above the Great Northern Highway between Munjina and Newman near 22°28′S, 118°42′E, showing the undulating Hamersley Surface with duricrust continuous on more elevated parts but disintegrating on the slopes and removed from valleys. (b) Typical undulating Hamersley Surface capped by a partly eroded duricrust formed on strata of the Hamersley Group, Hamersley Range.

Morris and Ramanaidou (2007) concluded that maximum accumulation of CID occurred in the middle Miocene during and following the Miocene climatic optimum, which is dated at *ca* 17–14 Ma (Bradley, 2014; Zan, Fang, Yan, Zhang, & Lu, 2015). Combined (U–Th)/He and $^{4}He/^{3}He$ dating of late-stage authigenic goethite from the CID at Yandicoogina gave early to late Miocene ages of *ca* 18–5 Ma (Heim, Vasconcelos, Shuster, Farley, & Broadbent, 2006) (employing the time-scale of Gradstein, Ogg, & Smith, 2004), although Morris, Kneeshaw, and Ramanaidou (2007) queried aspects of the sampling strategy and statistical treatment of the data. (U–Th)/He chronology of hematitic nuclei of ooids from a 32.8 m-deep diamond drill core sample of CID at Mesa J in the Robe Valley ESE of Deepdale yielded ages ranging from 25.7 ± 0.6 to 13 ± 0.4 Ma, with a central age of 18.3 ± 3.5 Ma (Danišík et al., 2013). Because of possible loss of radiogenic He, for which no correction was applied, they concluded (p. 156) that 'the ages could be biased towards the younger values by an unknown factor.' The results imply that the hematitic nuclei of the dated samples may have formed as early as the late Oligocene. Goethitic rims yielded an early late Miocene age of 11.6 ± 3.0 Ma, and ferruginised wood fragments yielded late Miocene ages of 8.8 ± 0.6 and 7.5 ± 0.5 Ma (Danišík et al., 2013), but these ages also may be minima.

Global climates warmed during the late Oligocene–early Miocene (Macphail & Stone, 2004; Zachos, Pagani, Sloan, Thomas, & Billups, 2001), with the late Oligocene global mean annual air temperature 7.5°C higher than that in pre-industrial times (Lunt, Ross, Hopley, & Valdes, 2007). Minor increase in siliciclastic deposition in the offshore Carnarvon Basin implies increased precipitation in the Pilbara region during the late Oligocene and early Miocene (Macphail & Stone, 2004). The early Miocene climate in central and western Australia was warm and humid with patches of rainforest in the Fortescue Valley area (Martin, 2006). The (U–Th)/He chronology and the late Oligocene–early Miocene environment are consistent with lateritic weathering in the source area under warm and humid conditions at that time, with CID deposition and precipitation of secondary goethite during the middle to late Miocene. By the mid-Miocene, regular flows in the Western Australian paleodrainage systems had stopped, marking the first major step towards aridity (Martin, 2006). According to Ramanaidou and Morris (2010), the stripping phase of the Hamersley Surface culminated at *ca* 12 Ma, with deposition of

calcreted limestone of the Oakover Formation indicating increasingly cold, arid conditions prior to the onset of Pliocene warming.

CID are present along tributaries of the Fortescue River upstream of Gregory Gorge but absent along the Fortescue River and its tributaries downstream from that point (Figure 1). During the Miocene, the Fortescue River evidently had its outlet to the sea via the paleo-Robe River and subsequently has taken a northwestward course from Gregory Gorge. Hence, the Robe is a beheaded stream.

Paleomagnetic study

Sampling sites

A ferruginous weathering zone that is developed on strata of the Hamersley Group (Figure 3) was sampled within an estimated 15 m below the duricrusted Hamersley Surface at 17 sites in roadcuts along the Great Northern Highway between Munjina and Newman (Figure 4, sites HM01–11, HM15–20) and at three sites in exposures along the adjoining Karijini Drive (sites HM12–14). Two to eight small (kg) block samples were collected at each site (total 91 samples) and oriented using a sun compass or a magnetic compass, or both. Site and sample details are given in Appendix 1.

Laboratory techniques

From three to seven cylindrical specimens 2.2 cm high × 2.5 cm diameter were machined from each of the 91 block samples, and at least two specimens per block were processed by routine paleomagnetic laboratory methods (Butler, 1992; Collinson, 1983) at CSIRO, North Ryde, NSW. These specimens were subjected to stepwise thermal demagnetisation with 10–15 steps up to 680°C, using a Magnetic Measurements MMTD80 shielded furnace. Remanent magnetism was measured using a 2G 755R three-axis cryogenic magnetometer.

Components of magnetism were isolated using an interactive version of Linefind (Kent, Briden, & Mardia, 1983), in which linear segments are fitted to data points weighted according to the inverse of their measured variances. The highest temperature to which any particular component remains stable, i.e. the highest unblocking temperature observed, was generally taken as an indication of the Curie temperature of the mineral carrying that component, bearing in mind that a particular unblocking-temperature spectrum sometimes may not extend as high as the Curie temperature.

Results

Results for the ferruginous weathering zone are given in Table 1. Figure 5 shows orthogonal projections of end-points of natural remanent magnetism and after stepwise thermal demagnetisation, and Figure 6 shows *in situ* directions for the weathering zone. Both normal and reversed polarities are present, specimens typically showing normal, reversed or mixed components of magnetism. Specimens such as HM01g1 and HM03b2 clearly show both polarities, consistent with these specimens being magnetised over two polarity chrons (time interval of say 10^5–10^6 years). This is an indication that geomagnetic secular variation is largely averaged out at the sample level, suggesting that the mean pole position calculated is a reliable paleomagnetic pole. Indeed, the mixed components of both polarities in some specimens (e.g. Hm03d1) could not be separated by either thermal or alternating-field demagnetisation methods. Consequently, many of the specimens treated did not yield useful results. A reversal test (McFadden & McElhinny, 1990) was conducted on the



Figure 3. Sampling site HM05 in a roadcut on the Great Northern Highway 8.8 km by road south of Munjina, showing the ferruginised weathering zone beneath the duricrusted Hamersley Surface developed on the Mount McRae Shale, Hamersley Group. Viewer looking northeast.

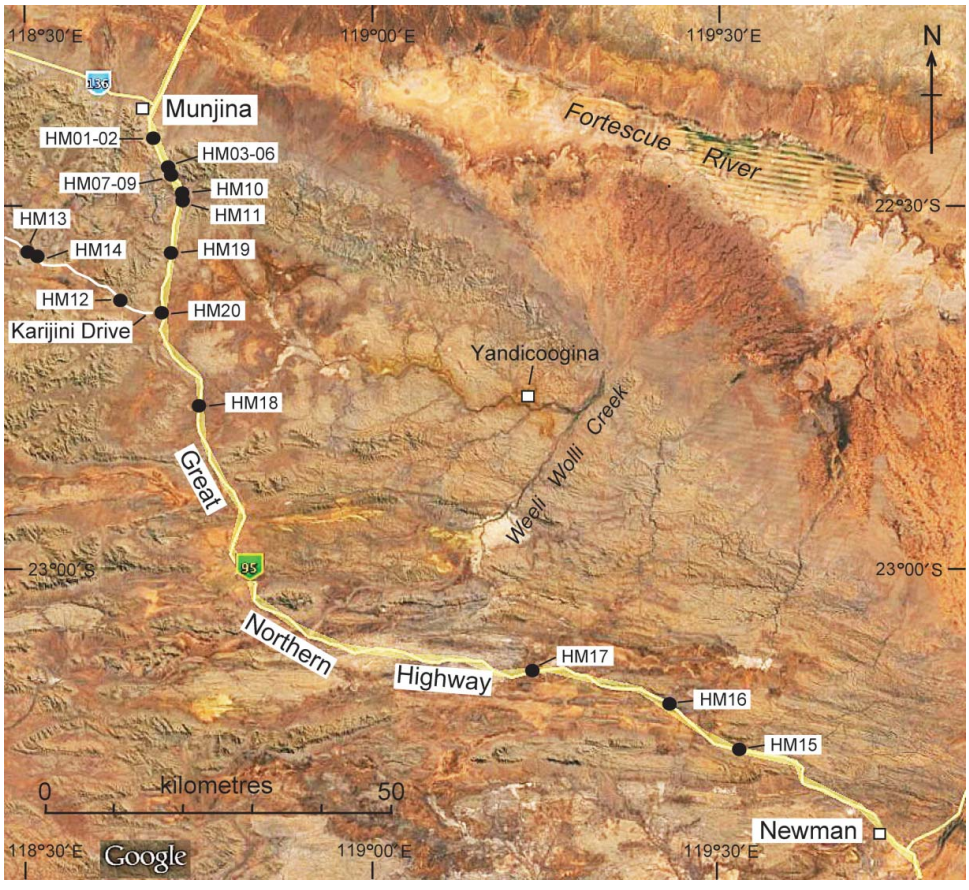


Figure 4. Map of sampling sites along the Great Northern Highway between Munjina and Newman and the adjoining Karijini Drive. Site details are given in Appendix 1. Background is a Google Earth image.

two anti-parallel components of declination $D = 359.2°$, inclination $I = -54.3°$ ($n = 23$, $r = 22.205$, $k = 27.659$ and $\alpha_{95} = 5.9°$) and $D = 183.9°$, $I = 51.9°$ ($n = 32$, $r = 30.958$, $k = 29.736$ and $\alpha_{95} = 4.7°$). The groupings share a similar precision (dispersion), and the reversal test proved positive, with the angle between the two groups ($\gamma_o = 3.7°$) being less than the critical angle ($\gamma_c = 7.7°$) at the $p = 0.05$ probability level. Angular resolution is better than 10° but not less than 5°, which means that while positive, the reversal test is B class.

Use of the specimen mean, rather than the site mean, is justified on the basis that individual specimens blocked in slowly enough to internally average secular variation, based on the observations above. A stable, high-temperature (680°C) component carried by hematite was revealed, with a mean direction ($n = 55$ specimens) $D = 182.0°$, $I = 52.9°$ ($\alpha_{95} = 3.6°$), indicating a pole position at latitude $\lambda_p = 77.6°$S, longitude $\varphi_p = 113.2°$E ($A_{95} = 4.3°$) and paleolatitude $\lambda = 33.5 +3.6/-3.3°$S.

Table 1. Paleomagnetic summary for the ferruginous weathering zone beneath the Hamersley Surface.

Site	n	D_h (°)	I_h (°)	k	α_{95} (°)
HM01	10	180.7	48.9	55.5	6.5
HM02	3	356.3	−60.5	16.6	31.3
HM03/4	4	196.1	55.9	24.1	19.1
HM05	6	3.6	−40.8	94.9	6.9
HM07	6	175.4	61.9	26.0	13.4
HM10/11	6	1.0	−46.3	42.9	10.3
HM12+20	5	169.1	57.1	90.7	8.1
HM13/14	5	186.2	44.0	57.7	10.2
HM15	3	193.1	64.9	254	7.8
HM17/18	7	3.7	−59.2	15.8	15.7
Mean	$N = 10$	182.5	54.2	73.9	5.7
	$n = 55$	182.0	52.9	29.0	3.6

Mean pole positions:
sites: $N = 10$, $\lambda_p = 76.9°$S, $\varphi_p = 110.7°$E, $K = 62.8$, $A_{95} = 6.1°$;
specimens: $n = 55$, $\lambda_p = 77.6°$S, $\varphi_p = 113.2°$E, $K = 20.5$, $A_{95} = 4.3°$.
Polarities shown reflect the predominant polarities of each site; N/n, number of sites/specimens showing ChRM (characteristic remanent magnetism); D_h and I_h, *in situ* declination and inclination; λ_p and φ_p, latitude and longitude; k and K, precision parameters; α_{95} and A_{95}, the half-angle of the 95 percent confidence cone for the mean directions and poles, respectively (Fisher, 1953).

Paleomagnetic age of the weathering zone

The paleomagnetic age of the ferruginous weathering immediately beneath the Hamersley Surface can be determined by comparing the indicated paleomagnetic pole position at 77.6°S, 113.2°E ($A_{95} = 4.3°$) with paleomagnetic poles for dated Cenozoic rocks in eastern Australia (Schmidt & Embleton, 1976). The early work by Irving and Green (1958) clearly showed that Australia's break-up from Antarctica is recorded by a regular shallowing of paleomagnetic inclinations throughout much of the Cenozoic as Australia drifted from near the South Pole to lower latitudes. Later work was aimed at refining the Cenozoic pole path to enable more accurate

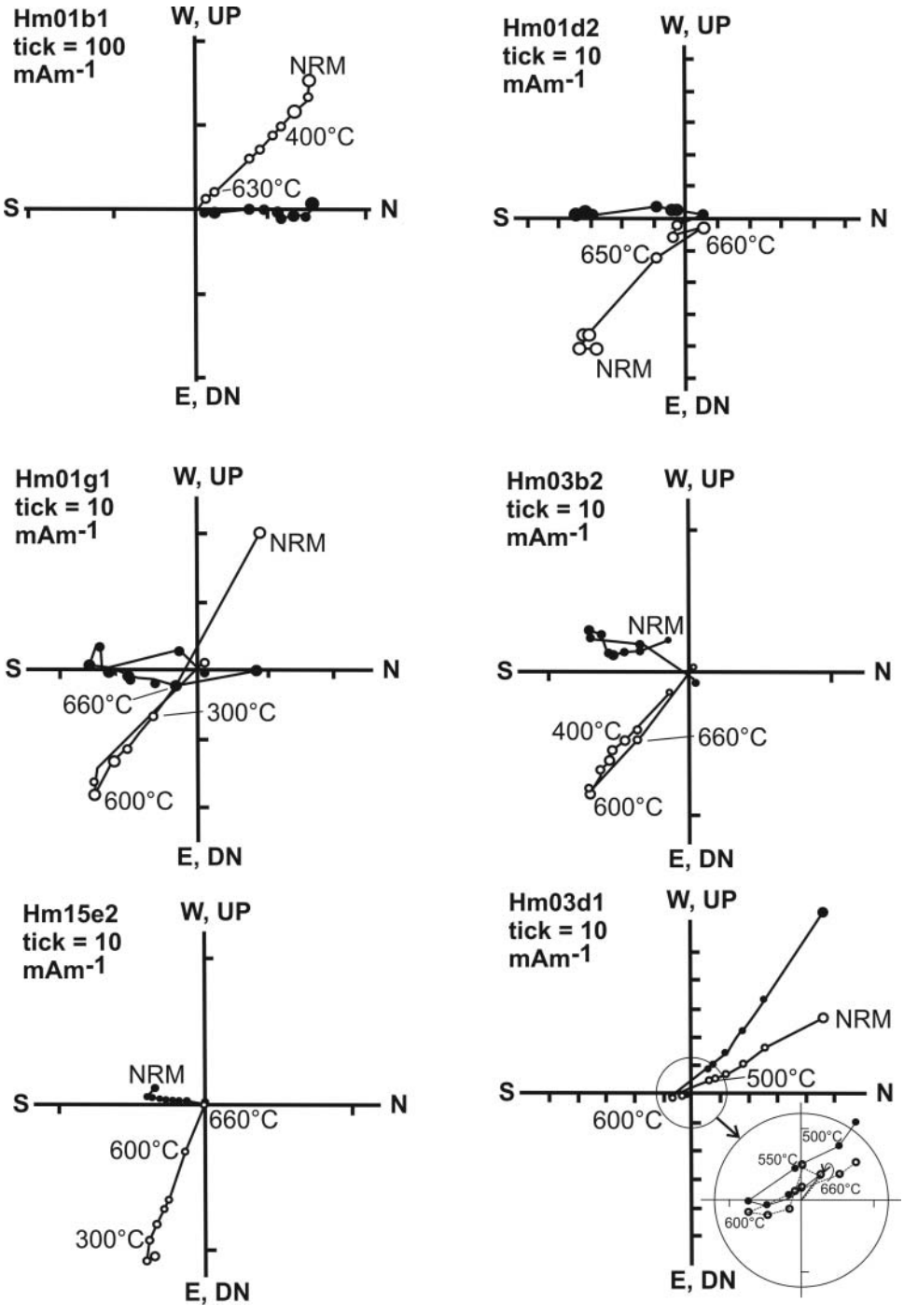


Figure 5. Orthogonal projections of remanence vectors for specimens from the weathering zone beneath the Hamersley Surface after stepwise thermal demagnetisation up to 680 °C. Solid circles plot on the horizontal plane and open circles on the vertical plane (DN, down).

paleogeographic reconstructions. Wellman, McElhinny, and McDougall (1969) obtained a pole at 78.5°S, 84.3°E (A_{95} = 4.5°) for 34 lava flows from the Nandewar Volcano, dated by K–Ar at 17.5 ± 0.3 Ma. Both normal and reversed polarities are present in a 350 m section, and Wellman et al. (1969) thought that the data spanned sufficient time to average out secular variation. Wellman (1975) determined a pole at 74.2°S, 114.8°E (A_{95} = 4.1°) for 37 lava flows from the Tweed Volcano succession, dated by K–Ar at 22 ± 2 Ma; a 700 m section showed both normal and reversed polarities and four changes in polarity.

Subsequent to establishing this 'calibrated' Cenozoic pole path, several studies sought to utilise the path to date weathering horizons many of which are prominent in the Australian landscape. Schmidt and Embleton (1976) studied the now-dissected peneplain of the North Perth Basin and assigned a late Oligocene to early Miocene age for the deep weathering of that area. Idnurm and Senior (1978) found two ages, latest

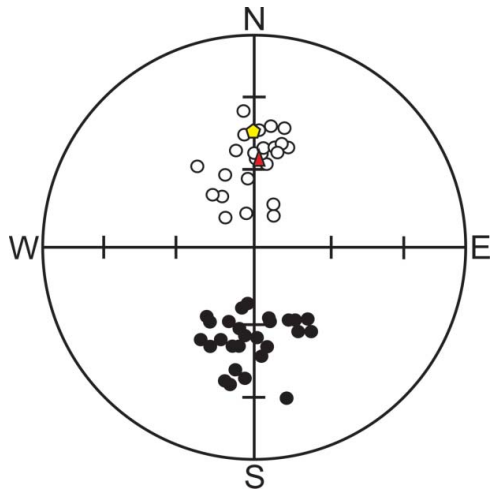


Figure 6. Stereographic projection of characteristic remanent magnetisation (ChRM) *in situ* directions for the weathering zone beneath the Hamersley Surface isolated by stepwise thermal demagnetisation. Solid circles plot on the lower hemisphere and open circles on the upper hemisphere. The present field is shown by the red triangle and the dipole field by the yellow pentagon.

Cretaceous (Maastrichtian) to early Eocene and late Oligocene, for the Morney and Canaway weathering profiles, respectively, in the Eromanga and Surat basins of southern Queensland. Paleomagnetic evidence for the Cretaceous weathering event was later found in the New England area of northern NSW (Schmidt & Ollier, 1988) several hundred kilometres south of the Queensland area studied by Idnurm and Senior (1978).

Idnurm (1985) revised the late Mesozoic–Cenozoic apparent polar wander path (APWP) for Australia using well-dated sedimentary successions from western Victoria and a drill core from northern Western Australia. He pointed out that the previous pole path, although arguably based on well-averaged paleomagnetic directions from volcanic piles in eastern Australia, nevertheless displayed 'zig-zag irregularities.' While the K–Ar ages were valuable, Idnurm (1985, 1994) cast doubt on the accuracy of the pole positions and suggested that many of the flows are serially correlated. That is, rather than the volcanic pile being evenly distributed through time, many flows may occur within a short period separated by long hiatuses. This results in secular variation being under-sampled and an APWP without an expected smooth and regular form.

Musgrave (1989) introduced a weighted least-squares approach combining the poles from Cenozoic volcanic rocks with those from weathering profiles, and concluded that the data for the volcanics are not unreliable. This allowed the radiometric ages to calibrate the APWP that can be used to determine the age of undated poles, with age being a function of distance along the path. Musgrave's APWP also differs from Idnurm's path in not requiring a variable rate of polar wander throughout the Cenozoic. Thus, Musgrave's path is actually more consistent with implied hotspot 'poles' (Duncan, 1981), which for the Cenozoic are in good agreement. More recent analyses of the Pacific and Australian plate motions have been undertaken (e.g. Knesel, Cohen, Vasconcelos, & Thiede, 2008) but the rotation parameters have not been tabulated to compare with those of Duncan (1981). The most comprehensive analysis of plate motions to date is arguably that of Doubrovine, Steinberger, and Torsvik (2012), who investigated the effects of moving hot spots and developed the global moving hot spot reference frame. For times younger than 50 Ma the motions are essentially the same as for Torsvik, Müller, Van der Voo, Steinberger, and Gaina (2008), which we use below. Thus, for late Cenozoic times, a fixed or moving hot spot frame makes very little difference. Doubrovine et al. (2012, table 4) shows that the net rotation of the lithosphere is less than 0.2°/Ma back to 50 Ma. For 20 Ma events, the maximum error we might expect is 4° in comparing the North American and Australian pole paths, which is less than the A_{95} confidence limit of the Australian pole positions.

We have compared the pole position for the weathering zone beneath the Hamersley Surface with Idnurm's poles (Figure 7a), Musgrave's APWP (Figure 7b) and the hotspot 'poles' given by Duncan (1981) (Figure 7c). To better quantify our comparison we have adopted the reduced chi-square method developed by Gordon and Cox (1980) (see also Acton & Kettles, 1996). The insets in Figure 7 show the reduced chi-square values plotted against time; time is inferred by linear interpolation between the individual dated poles for each data set (Supplementary Paper). The maximum likelihood ages indicated for the weathering-zone pole are: Idnurm's poles, 10 Ma; Musgrave's APWP, 23 Ma; and the hotspot 'poles,' 22 Ma. Thus, there is good agreement between results based on Musgrave's and the hotspot analyses.

One further comparison involves transfer of the North American late Mesozoic–Cenozoic poles (Torsvik et al., 2008) to Australia and chi-square calculations. The North American database is about an order of magnitude larger than that of Australia, so, notwithstanding cumulative errors, which may invalidate the late Mesozoic data, the Cenozoic data should provide a reasonably robust path to compare with the Hamersley weathering-zone pole. The results of this comparison are plotted in Figure 7d and give a maximum likelihood age of 28 Ma. Rejecting the age estimate from Idnurm's poles as an outlier leaves three estimated ages of 23, 22 and 28 Ma, with an average of *ca* 24 ± 3 Ma, for the formation of hematite in the ferruginous zone immediately beneath the Hamersley Surface. We can only speculate as to why Idnurm's poles appear to be anomalous. While his studies concentrated on sediments to overcome the poor averaging of secular variation that afflicts the poles derived from eastern Australian Cenozoic volcanics, the magnetisation processes of sediments can be complicated. Indeed, Idnurm acknowledged this when using magnetisations of intermediate stability, because he could not isolate the most stable magnetisations using paleomagnetic methods (Idnurm, 1985, 1994). Components of intermediate stability can be contaminated with some other components of unknown source. Our comparisons herein incorporate results from sediments, volcanics and weathering horizons alike.

Our results supersede preliminary findings for untreated specimens from the same ferruginous zone (Schmidt &

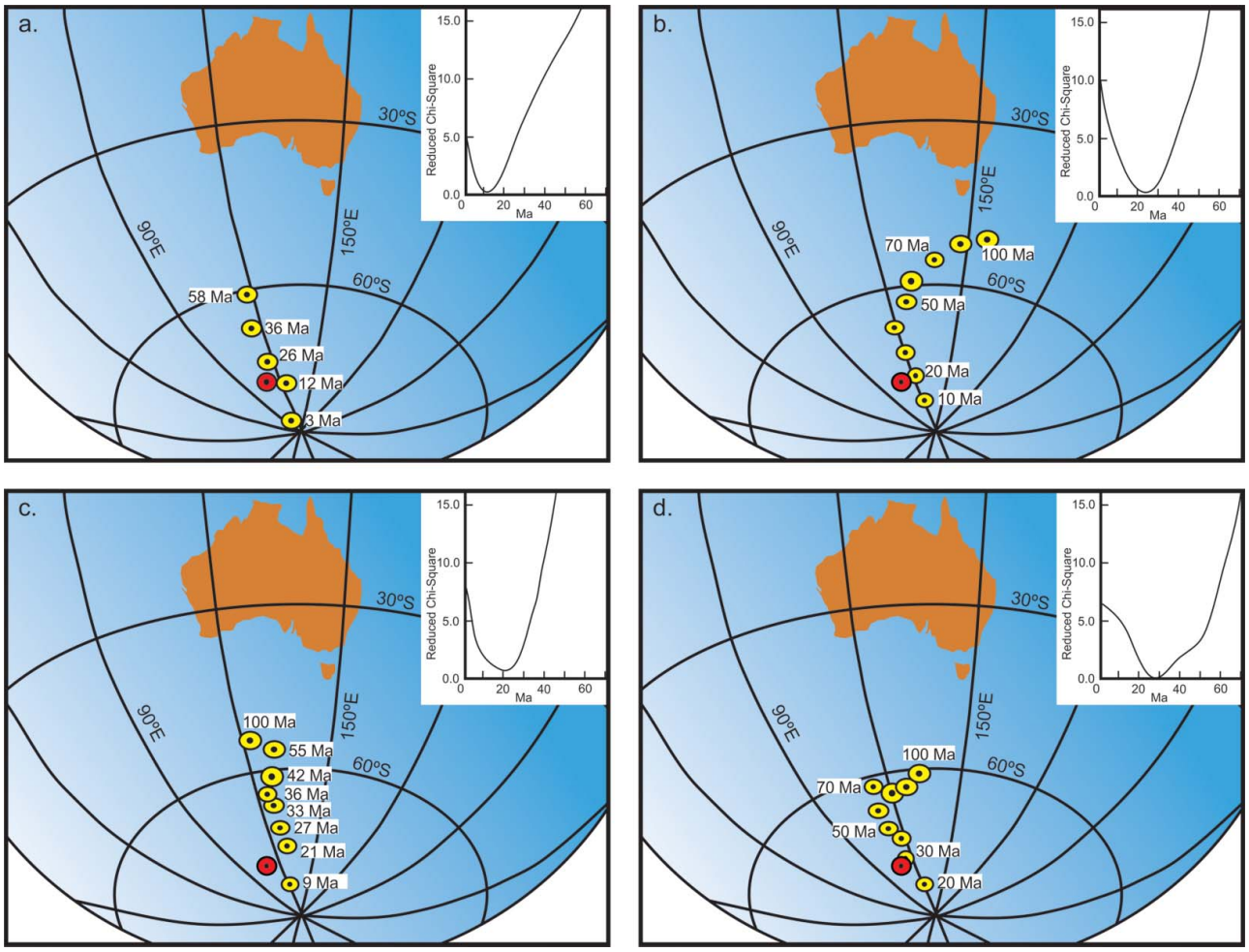


Figure 7. Cenozoic APWP for Australia, from (a) Idnurm (1985, 1994), (b) Musgrave (1989), (c) hotspot 'poles' (Duncan, 1981), and (d) rotated North American path (Torsvik et al., 2008). The red dots represent the pole for the weathering zone beneath the Hamersley Surface. The insets show the reduced chi-square values comparing the Hamersley Surface pole with poles from respective Cenozoic data sets.

Williams, 2002). The paleomagnetic pole determined herein is similar to that determined by Abrajevitch, Pillans, and Roberts (2014) for hematitic uniform pigment in a decorative 'Print Stone' developed on the Mount McRae Shale approximately 60 km northwest of Tom Price; they concluded that the pigment was acquired through percolation of meteoric waters enriched in iron during the late Oligocene–early Miocene. Our result also is in agreement with paleomagnetic data indicating a late Oligocene to early Miocene age (25–20 Ma) for laterisation in the Perth Basin (Schmidt & Embleton, 1976).

Conclusions

A ferruginous weathering zone immediately beneath the Hamersley Surface, developed on strata of the Hamersley Group, was sampled at 20 sites along the Great Northern Highway between Munjina and Newman and the adjoining Karijini Drive. Stepwise thermal demagnetisation revealed a stable, high-temperature (680°C) component carried by hematite, with a mean direction (n = 55 specimens) D = 182.0°, I = 52.9° (α_{95} = 3.6°), indicating a pole position at 77.6°S, 113.2°E (A_{95} = 4.3°) and a paleolatitude λ = 33.5 +3.6/–3.3°S. The presence of both normal and reversed polarities indicates that the remanent magnetism was acquired over an extended time interval (10^5–10^6 years).

Chi-square tests on the pole position for the weathering zone and three different sets of Cenozoic poles, viz. those for dated volcanic rocks in eastern Australia and for Australian Cenozoic weathering horizons (Musgrave, 1989), and inferred poles from Pacific Ocean and Indian Ocean hotspot analyses (Duncan, 1981) and North American Cenozoic poles (Torsvik et al., 2008) rotated to Australian coordinates, yielded an age of *ca* 24 ± 3 Ma, i.e. late Oligocene to early Miocene, for the weathering-zone pole. This is interpreted as the time of formation of hematite in the sampled ferruginous zone.

The late Oligocene–early Miocene age for the sampled weathering zone implies that it is related to the HS3 weathering episode of Morris and Ramanaidou (2007). The results are consistent with this weathering episode yielding the hematitic nuclei of CID ooids with (U–Th)/He ages ranging from 25.7 ± 0.6 to 13 ± 0.4 Ma (Danišík et al., 2013), and with the conclusion that detritus derived from the erosion of a lateritic regolith beneath the Hamersley Surface was incorporated in CID during

the Miocene (Morris & Ramanaidou, 2007; Ramanaidou & Morris, 2010).

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No potential conflict of interest was reported by the authors.

Supplementary paper

The spreadsheet used to combine Euler poles and perform rotations and other calculations.

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Appendix 1

Sampling sites

Small (kg), oriented block samples were collected from a reddened, laterised zone developed on strata of the Hamersley Group within an estimated 15 m below the duricrusted Hamersley Surface. The indicated formations are taken from the Roy Hill 1:250 000 geological map (Thorne & Tyler, 1996) and the Newman 1:250 000 geological map (Tyler, Williams, & Hunter, 1990).

Sites HM01, HM02. Road cut, east side of the Great Northern Highway, at 22°24.68′S, 118°41.14′E. Ferruginous weathering developed on fine-grained clastic sediments and chert of the Wittenoom Formation (Roy Hill 1:250 000 geological map).

Sites HM03, HM04. Road cut, west side of the Great Northern Highway, at 22°27.10′S, 118°42.19′E. Ferruginous weathering developed on the Wittenoom Formation (Roy Hill 1:250 000 geological map).

Sites HM05, HM06. Road cut, west side of the Great Northern Highway, at 22°27.24′S, 118°42.27′E. Ferruginous weathering developed on fine-grained clastic sediments of the Mount McRae Shale (Roy Hill 1: 1:250 000 geological map).

Sites HM07, HM08. Road cut, east side of the Great Northern Highway, at 22°27.49′S, 118°42.35′E. Ferruginous weathering developed on the Mount McRea Shale (Roy Hill 1:250 000 geological map).

Site HM09. Road cut, west side of the Great Northern Highway ≈ 200 m south of sites HM07–08, at 22°27.76′S, 118°42.45′E. Ferruginous weathering developed on the Mount McRea Shale (Roy Hill 1:250 000 geological map).

Site HM10. Road cut adjacent to the Great Northern Highway, at 22°29.25′S, 118°43.89′E. Ferruginous weathering developed on the Brockman Iron Formation (Roy Hill 1:250 000 geological map).

Site HM11. Road cut ≈ 150 m south of site HM10 adjacent to the Great Northern Highway, at 22°29.30′S, 118°43.89′E. Ferruginous weathering developed on the Brockman Iron Formation (Roy Hill 1:250 000 geological map).

Site HM12. Exposure by Karijini Drive, at 22°38.20′S, 118°38.43′E. Ferruginous weathering developed on the Mount McRea Shale (Roy Hill 1:250 000 geological map).

Site HM13. Exposure by Karijini Drive, at 22°34.91′S, 118°30.89′E. Ferruginous weathering developed on the Mount McRea Shale (Roy Hill 1:250 000 geological map).

Site HM14. Exposure by Karijini Drive ≈ 600 m east of site HM13, at 22°34.97′S, 118°31.09′E. Ferruginous weathering developed on the Mount McRea Shale (Roy Hill 1:250 000 geological map).

Site HM15. Road cut, north side of Great Northern Highway, at 23°14.24′S, 119°31.63′E. Ferruginous weathering developed on iron-formation and dolerite of the Weeli Wolli Formation (Newman 1:250 000 geological map).

Site HM16. Road cut, north side of Great Northern Highway, at 23°11.14′S, 119°26.53′E. Ferruginous weathering developed on the Mount McRae Shale (Newman 1:250 000 geological map).

Site HM17. Road cut, Great Northern Highway, at 23°08.05′S, 119°14.01′E. Ferruginous weathering developed on the Marra Mamba Iron Formation (Newman 1:250 000 geological map).

Site HM18. Road cut, Great Northern Highway, at 22°46.00′S, 118°45.38′E. Ferruginous weathering developed on the Brockman Iron Formation (Roy Hill 1:250 000 geological map).

Site HM19. Road cut, west side of the Great Northern Highway, at 22°34.23′S, 118°43.15′E. Ferruginous weathering developed on the Brockman Iron Formation (Roy Hill 1:250 000 geological map).

Site HM20. Road cut, west side of Great Northern Highway, at 22°39.13′S, 118°42.44′E. Ferruginous weathering developed on the Brockman Iron Formation (Roy Hill 1:250 000 geological map).